UK Train Rides: Team Project Plan

End-to-End Analytics, Modeling, SQL Data Pipeline & Dashboard Implementation

STATUS: FIXED VERSION | TARGET OPERATIONAL ROADMAP

1. Team Roles & Phase Ownership

The table below details the responsibility matrix across all six project phases. Each member is assigned specific leads, while collaborating on shared deliverables.

Person	Phase & Assigned Role
Everyone	Phase 1: Business understanding
Mohamed, Lamis	Phase 2: Data understanding (EDA)
George, Sondos	Phase 3: Data preparation
Lamis, Mohamed	Phase 4: SQL data model and analysis questions
Nour	Phase 5: Modeling and evaluation
Omar	Phase 6: Deployment, review, final report, presentation

Phase 1: Business Understanding and Domain Knowledge (Everyone)

1.1 The Problem

A UK train operating company has ticket sales data and wants to understand its demand and revenue and forecast next month's rides, daily revenue, and demand per ticket class. Your job is to turn that data into decisions managers can use.

1.2 How the Business Works

- Train operating companies run the trains. Network Rail owns the tracks and most stations. Your data most likely comes from one operator's ticket sales.
- Costs are mostly fixed, so a train costs about the same full or empty. The goal is full trains at good prices, which is why prices vary.

1.3 Domain Knowledge

These are typical industry facts. Verify each one against the real dataset.

Topic	What to Know
Ticket types	Advance (cheapest, fixed train, bought early), Off-Peak (flexible, outside rush hours), Anytime (flexible, most expensive), Season (commuter passes). Also Single vs Return.
Ticket class	Standard (usually the majority of tickets) or First (fewer tickets, high price, can matter a lot for revenue).
Pricing	Peak hours cost more, longer distance costs more, and Advance prices rise as the date gets closer (demand-based pricing).
Railcard	A discount card (about one-third off), so the price paid is lower than the full fare.
Purchase channel	Online or app, ticket machine, or ticket office. Online is typically the largest.
Journey status	On Time, Delayed, Cancelled. Delayed or cancelled journeys can lead to refunds, which reduce real revenue.
Demand drivers	Weekday vs weekend, rush hours, month and season, bank holidays, strikes, events, and weather.

1.4 Key Measures

Measure	Meaning & Business Significance
Demand	Number of rides or tickets sold
Revenue	Total money from tickets (rides × average price)
Yield	Average revenue per ride
Punctuality	% of trains on time
Refund rate	% of tickets refunded

> **KEY PRINCIPLE:** Revenue grows from more rides or higher yield. Always check which factor caused a change during period-over-period analysis.

1.5 Stakeholders and Decisions

Who	Decides	Needs from You
Operations	How many trains to run	Busy routes and times, ride forecast
Pricing	Ticket prices	Revenue and demand by type and class
Finance	Budget	Daily revenue forecast
Customer service	Delay and refund handling	Delay and refund patterns

1.6 Business Questions

- Which routes, ticket types, and classes earn the most?
- Which days, hours, and months are busiest?
- How often are trains delayed, and how much revenue is lost to refunds?
- Required forecasts: next month's rides, daily revenue, and demand per ticket class.

1.7 Success, Assumptions, and Risks

- **Success Criteria:** Managers get answers from the dashboard alone, forecasts beat a simple baseline (e.g., "same as last week"), and every decision is documented.
- **Core Assumptions:** Each row represents one ticket, prices are in British Pounds (£), and historical patterns continue into the forecast horizon.
- **Project Risks:** Short data duration, strikes or unusual external events, messy data, and team members cleaning data inconsistently.

1.8 Why This Matters for the Data Work

- Spots errors: First class cheaper than Standard on the same route is suspicious.
- Guides cleaning: a £0 ticket may be a legitimate refund, not an error.
- Creates features: peak flag, holiday flag, and days booked in advance come from domain knowledge.
- Makes insights useful: "Fridays at 5 pm are the peak, so add capacity" is a decision, not just a number.

1.9 Team Action

Each person reads this section, researches one topic from 1.3 in detail, and writes 5 to 6 sentences summarizing findings. Then merge all contributions into one shared baseline document.

Phase 2: Data Understanding / EDA (Mohamed and Lamis)

> **PHASE GOAL:** Deliverable: One comprehensive PDF report that fully explains the dataset, statistical properties, relationships, and ERD architecture.

Part A: Statistics, Visualizations, and Report Text

1. Load the data and record total rows, columns, and date range.
2. Build the data dictionary table: variable, data type, variable type, and a plain-language meaning for each column.
3. Numerical summary: mean, median, min, max, standard deviation, quartiles (df.describe()).
4. Categorical summary: counts and percentages (value_counts).
5. Quality report: missing values, duplicates, outliers, inconsistent text (report only, no fixing in this phase).
6. Charts: histograms, bar charts, box plots, monthly and daily trend lines.
7. Relationships: correlation heatmap, scatter plots, price by ticket class, crosstabs.
8. Write one-line insights under every chart and table.
9. Check the data period (start and end dates). If it covers only a few months, explicitly highlight this as seasonality cannot be fully learned from a short sample.

Part B: ERD Design

1. Design a star schema: one fact table plus dimension tables (Date, Station, Ticket Class, Ticket Type, Payment).
2. Mark primary keys and foreign keys clearly.
3. Draw the complete entity relationship diagram in draw.io or dbdiagram.io.
4. Write a short description of each table.

Joint Responsibility: Assemble everything into the final EDA PDF and write key findings (5 to 8 executive bullets). George and Sondos read this PDF first, because it tells them what needs to be fixed during Phase 3.

Phase 3: Data Preparation (George and Sondos)

> **CRITICAL RULE (DATA LEAKAGE PREVENTION):** Anything learned from the data (outlier limits, skewness transforms, scaling values, encodings) must be calculated on the training period ONLY, then applied to the test period. Otherwise future information leaks into training and validation scores look artificially inflated. Sondos and Nour must agree on the cutoff date for the train/test split before Sondos begins.

Strict Order of Work: cleaning → missing values → outliers → skewness → feature engineering → feature selection

George: Cleaning and Missing Values

1. Data Cleaning

- Find and remove duplicate rows (duplicated(), drop_duplicates()).
- Fix data types (dates to datetime, prices to numbers).
- Standardize text (trim spaces, fix capitalization, merge spelling variants).

- Remove impossible values (negative prices, arrival before departure).

2. Handling Missing Values

- Count missing values per column.
- Decide per column, and document justification:
  - Numerical: fill with median (or mean if symmetric).
  - Categorical: fill with mode or "Unknown."
  - Mostly empty column or unusable rows: drop.
- Check again afterwards (isna().sum() should match expected state).

Sondos: Outliers, Skewness, Feature Engineering

3. Handling Outliers

- Detect with box plots and the IQR rule (below Q1 − 1.5×IQR or above Q3 + 1.5×IQR).
- Decide for each: keep if real (a long first-class trip), cap (winsorize) if extreme, remove if it's an error.
- Never delete blindly; document the specific business reason.

4. Fix Skewness

- Check skewness (df.skew()) and histograms.
- For right-skewed columns (like price), apply a log transform (np.log1p) or square root.
- Re-plot distribution to confirm shape improvement.

5. Feature Engineering (Transformation and Scaling)

- Date features: day of week, month, hour, weekend flag, holiday flag.
- Business features: journey duration, delay in minutes, revenue per ride, days booked in advance.
- Encoding: convert categories to numbers (one-hot or label encoding).
- Scaling: StandardScaler or MinMaxScaler for models that require normalized inputs.
- Time-series table: daily aggregated rows with total rides, revenue, and rides per class, plus lag features (yesterday, last week). This table is critical for Nour in Phase 5.

George: Feature Selection (After Feature Engineering)

6. Feature Selection

- Remove columns that add no value: IDs, constants, near-duplicates of other features.
- Check correlation between features and eliminate highly redundant predictors.
- Retain columns directly useful for the three target forecasts (rides, revenue, class demand).
- Document exactly which features were kept or dropped, along with rationale.

> **PHASE 3 OUTPUTS:** One final cleaned and prepared dataset (CSV/Parquet), a reproducible preprocessing Jupyter Notebook, and a structured decision log file.

Phase 4: SQL Data Model and Analysis Questions (Lamis and Mohamed)

Both parts below are mandatory requirements specified in the project brief.

Part A: SQL Data Model

1. Turn the ERD from Phase 2 into SQL tables (CREATE TABLE statements with primary and foreign key constraints).
2. Load the cleaned dataset from Phase 3 into the structured relational database.
3. Test table relationships with a series of JOIN queries to confirm seamless connectivity.
4. Save clean SQL DDL/DML scripts and final ERD diagram for inclusion in the final report.

Part B: Analysis Questions and Answers

1. Take the business questions from Phase 1 and expand with every additional query the dataset can answer across revenue, demand, customer profiles, delays, and refunds.
2. Verify each question is answerable with available columns, dropping queries reliant on missing data.
3. Answer each validated question using SQL queries and/or pandas code.
4. Plot every answer using Matplotlib / Seaborn.
5. Write a concise one-line insight for each visualization, linking directly to a management decision.

> **PHASE 4 OUTPUTS:** Executable SQL scripts, loaded database instance, and a consolidated document containing all analysis questions, query outputs, charts, and decision-focused insights.

Phase 5: Modeling and Evaluation (Nour)

> **PRE-MODELING CHECK:** Before starting, evaluate total dataset temporal span. If the dataset spans only a few months, annual seasonal phenomena (such as Christmas or summer spikes) cannot be learned. The model will rely primarily on day-of-week patterns and overall trend. Explicitly cite this as a modeling limitation.

Modeling Workflow

1. Define the three target variables: (a) total rides next month, (b) daily revenue next month, and (c) demand per ticket class.
2. Apply a strict time-based split: train on older dates, test on the most recent period. Never shuffle time-series observations.
3. Build a simple baseline model (e.g., "same as last week" naive benchmark).
4. Train candidate models in scikit-learn: start with Linear Regression, then evaluate ensemble models (Random Forest, Gradient Boosting).
5. Check class balance (if converting targets to classification): adjust imbalanced targets using class weights or resampling applied strictly to the training fold.
6. Tune hyperparameters (e.g., tree depth, estimator count) using time-aware cross-validation.
7. Forecast next month day-by-day and aggregate daily predictions to produce the monthly forecast total.

Model Evaluation

1. Apply task-appropriate evaluation metrics:
   - Regression / Forecasting: MAE, RMSE, R^2, MAPE.
   - Classification (if applied): Accuracy, Precision, Recall, F1-Score, Confusion Matrix.
2. Compare every model directly against the naive baseline. Any model failing to outperform the baseline must be rejected.
3. Plot actual vs. predicted values across the test period.
4. Select the top-performing model and write a technical justification.
5. Document all model constraints and operational limitations (e.g., strikes, unusual weather, short temporal span).

> **PHASE 5 OUTPUTS:** Modeling notebook, clean forecast plots, comprehensive model comparison evaluation table, and a concise technical rationale document.

Phase 6: Deployment, Review, Final Report, and Presentation (Omar)

Live Dashboard Development

1. Connect Tableau (or Power BI) to the final database / data tables and forecasting output datasets.
2. Build dedicated visual components for each business question answered in Phases 4 and 5.
3. Incorporate interactive filters (Date Range, Route, Ticket Class) alongside clear titles, tooltips, and legends.
4. Position forecast visualization panels side-by-side with historical trends for direct visual comparison.
5. Conduct user testing simulating an executive manager inspecting the dashboard for the first time.

Project Review & Quality Checkpoint

- Does every team member utilize the identical, final version of the dataset?
- Are all dashboard numbers perfectly synchronized with figures reported in Python and SQL notebooks?
- Is every analytical, cleaning, and modeling decision documented with reasoning?

Final Report Structure

1. Executive Summary & Business Problem Definition (Phase 1)
2. Data Description, Dictionary & ERD Architecture (Phase 2)
3. Data Cleaning, Preprocessing & Feature Engineering Pipeline (Phase 3)
4. SQL Relational Model, Business Query Analysis & Findings (Phase 4)
5. Predictive Modeling, Forecasting Results & Evaluation (Phase 5)
6. Tableau Dashboard Screenshots & User Interaction Guide
7. Strategic Conclusions & Executive Recommendations
8. Project Limitations & Future Enhancements

Final Presentation

Deliver a sleek executive slide deck communicating the project journey, focusing on strategic insights, demand/revenue forecasts, and actionable operational recommendations.

Handoff Sequence & Operational Dependencies

To ensure a smooth transition between phases and prevent pipeline bottlenecks, team members must adhere to the handoff hierarchy defined below:

Order	Who	Prerequisite (Needs First)
1	Everyone (Phase 1)	Project brief
2	Mohamed and Lamis (Phase 2)	Raw dataset
3	George (Phase 3)	Phase 2 EDA PDF report
4	Sondos (Phase 3)	George's cleaned data + train/test cutoff date agreed with Nour
5	George (Feature Selection)	Sondos's engineered features
6	Lamis and Mohamed (Phase 4)	Phase 3 final prepared dataset
7	Nour (Phase 5)	Phase 3 prepared data & daily time-series table
8	Omar (Phase 6)	Phase 4 SQL outputs & Phase 5 forecasting outputs

> **PARALLEL WORK STRATEGY:** To avoid idle time: Nour can begin initial baseline script setup using a preliminary daily aggregation table. Omar can construct dashboard wireframes and layout containers on raw data, swapping in final database tables once Phase 4/5 deliverables finalize.

Deliverables Master Checklist

#	Deliverable Description	Owner(s)
1	Business understanding document	Everyone
2	EDA PDF (data dictionary, statistics, charts, ERD)	Mohamed, Lamis
3	Cleaned and prepared dataset	George, Sondos

4	Preprocessing notebook and decision log	George, Sondos
5	SQL data model (scripts and diagram)	Lamis
6	Analysis questions with answers and charts	Mohamed
7	Forecast plots (rides, daily revenue, class demand)	Nour
8	Tableau dashboard	Omar
9	Final report	Omar
10	Final presentation	Omar